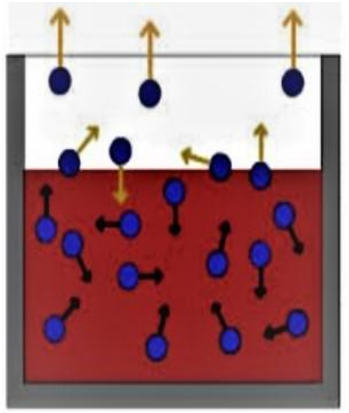
Fizika

9-sinf

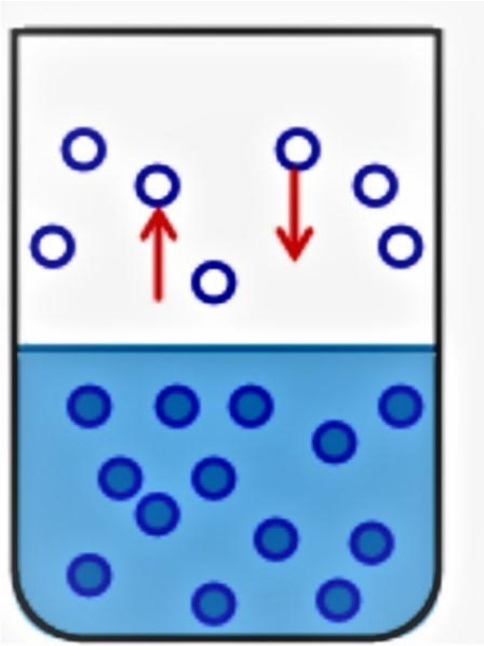
Mavzu: **Bug‘lanish va kondensatsiya.**



Bug‘lanish

Moddaning suyuq yoki qattiq agregat holatidan gaz holatga o‘tish jarayoni ***bug‘lanish*** deyiladi. Bug‘lanish tezligi bir necha omillarga bog‘liq:

- Temperaturaga;
- Shamolga;
- Suyuqlikning erkin sirtiga;
- Suyuqlik turiga.



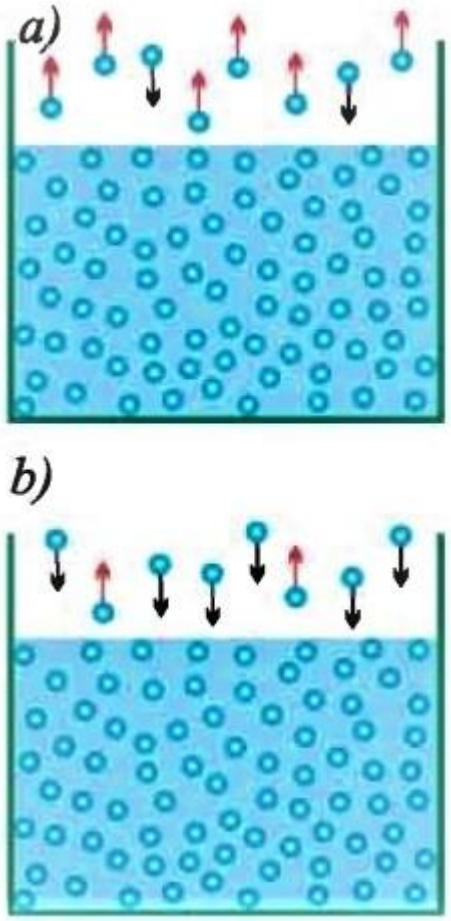
Solishtirma bug‘lanish issiqligi

O‘zgarmas temperaturada **1kg** suyuqlikni to‘la bug‘ga aylantirish uchun zarur bo‘lgan issiqlik miqdori ***solishtirma bug‘lanish issiqligi*** deyiladi va r harfi bilan belgilanadi.

$$r = \frac{Q_b}{m} \qquad Q_b = rm$$

Suv uchun solishtirma bug‘lanish issiqligi: $r = 2{,}3 \cdot 10^6 \text{J/kg}$

Kondensatsiya

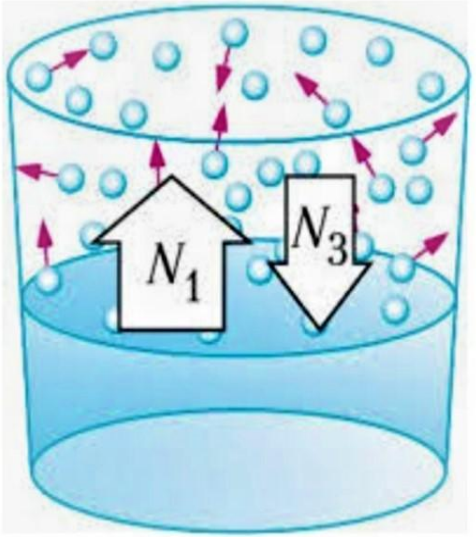


Bug‘ning suyuqlik holatiga o‘tish jarayoni ***kondensatsiya*** deb ataladi.

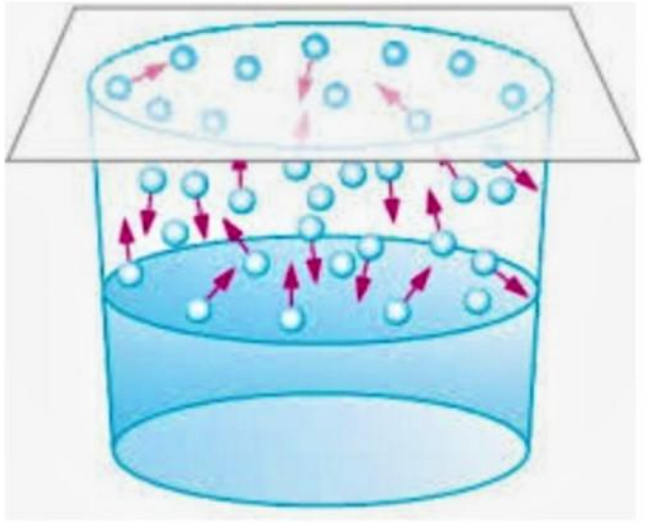
$$Q_k = -Q_b = -r \cdot m$$

To‘yingan va to‘yinmagan bug‘

O‘zining suyuqligi bilan dinamik muvozanatda bo‘lgan bug‘ ***to‘yingan bug‘*** deb ataladi. Bunday sharoitda suyuqlik ustida qaror topgan bug‘ning bosimi ***to‘yingan bug‘ning bosimi*** deyiladi.

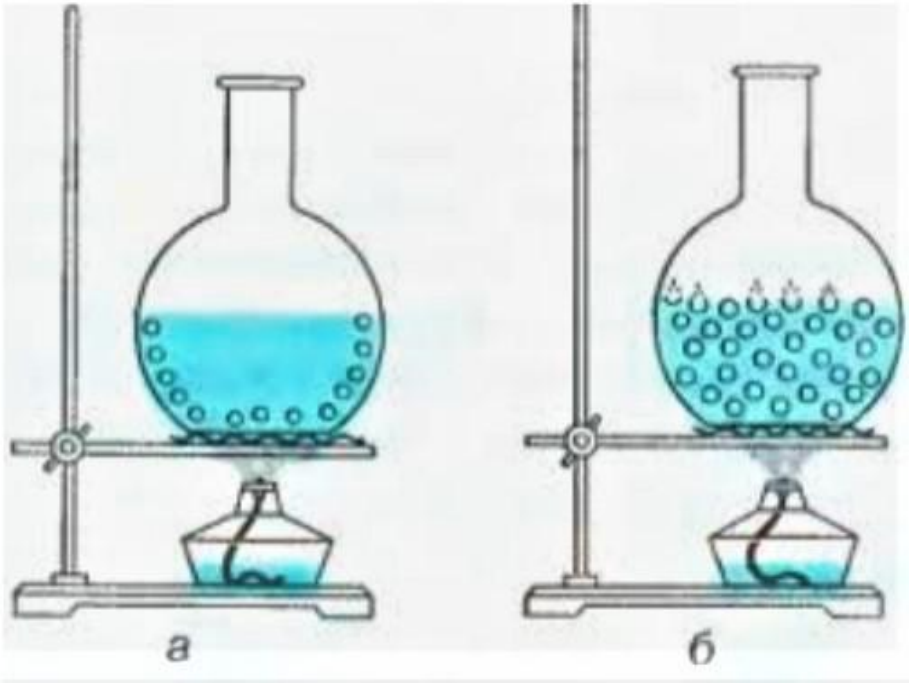


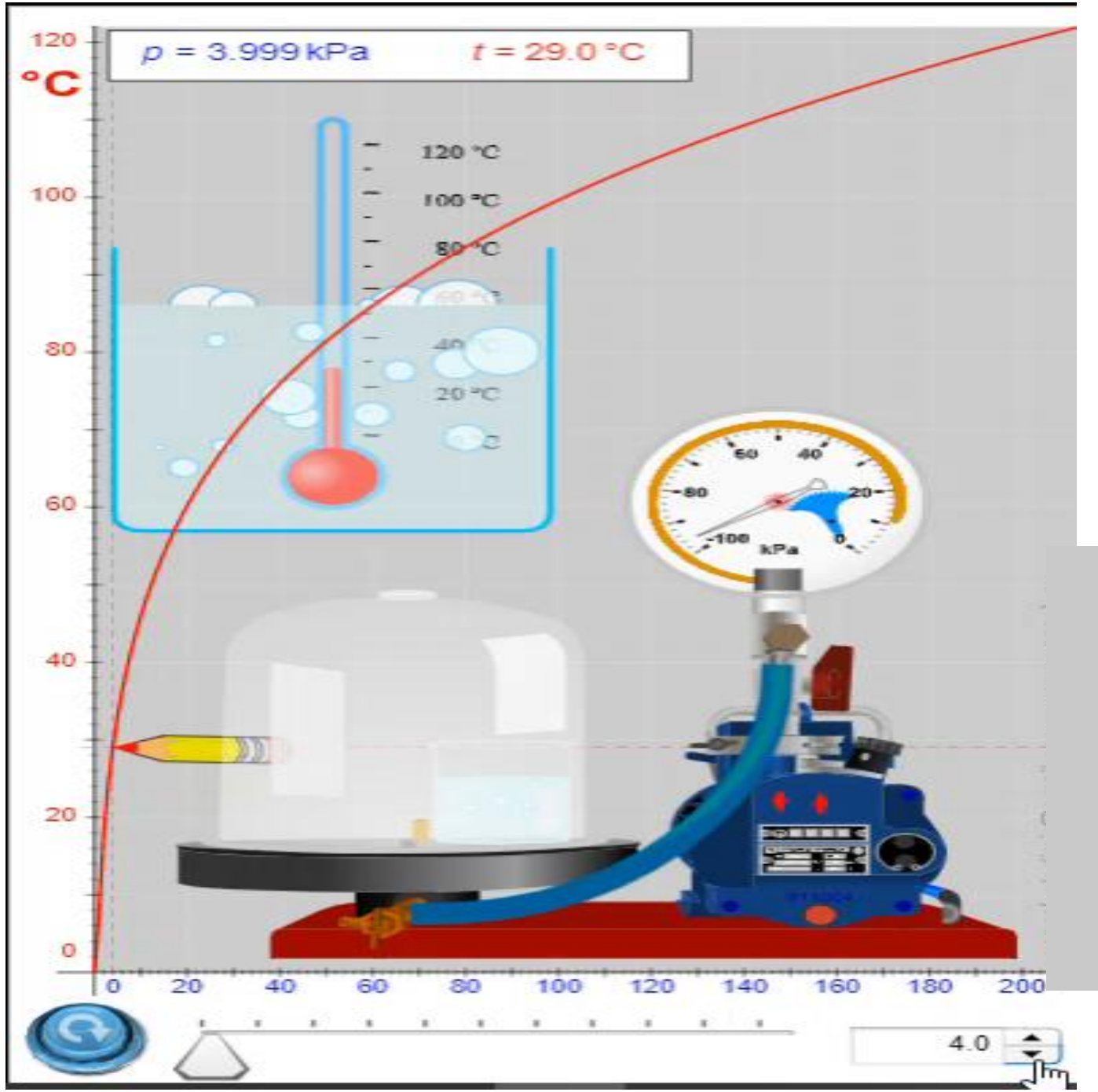
$$p = nkT$$

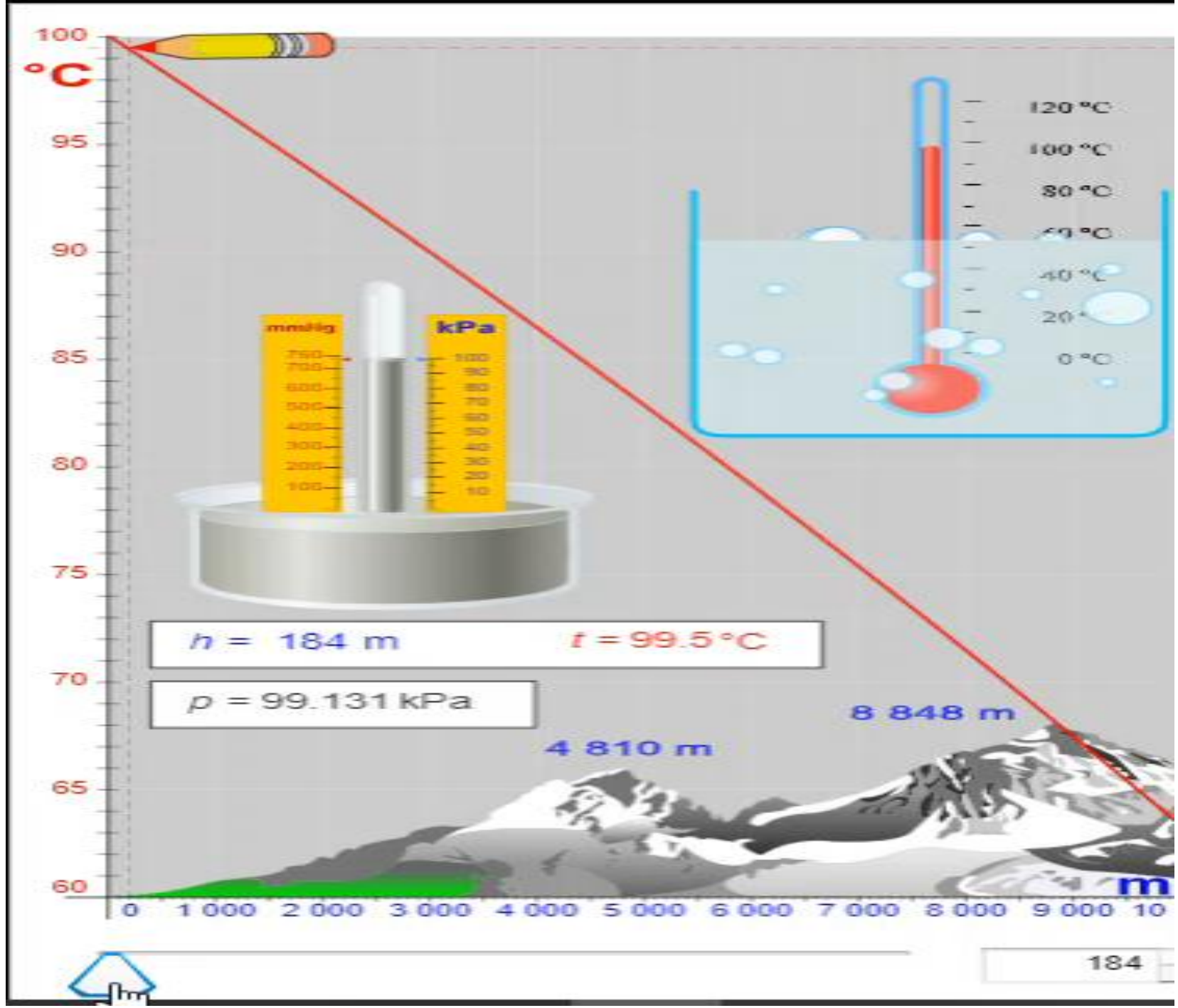


Qaynash

Suyuqlikning butun hajmi bo‘ylab bug‘ hosil bo‘lish jarayoni ***qaynash*** deb ataladi.




p = 3.999 kPa
t = 29.0 °C
°C
120
100
80
60
40
20
0
120 °C
100 °C
80 °C
40
20 °C
0
50
40
80
20
100
kPa
0
0 20 40 60 80 100 120 140 160 180 200
4.0


100
°C
95
90
85
80
75
70
65
60
mmHg
kPa
120 °C
100 °C
80 °C
40 °C
20 °C
0 °C
h = 184 m
t = 99.5 °C
p = 99.131 kPa
8 848 m
4 810 m
m
0
1 000
2 000
3 000
4 000
5 000
6 000
7 000
8 000
9 000
10
184

Masala

$\mathbf{0^0C}$ temperaturada olingan, massasi **10 g** bo‘lgan suvni qaynatish va bug‘ga aylantirish uchun qancha issiqlik miqdori kerak bo‘ladi?

Berilgan:

$t_1 = 0^0C$

$t_2 = 100^0C$

$1\ g = 0{,}001\ kg$

$r = 2{,}3 \cdot 10^6 \text{J/kg}$

$c = 4200\ \frac{J}{kg \cdot {}^0C}$

Formula:

$$Q = Q_q + Q_b$$

$$Q_q = cm(t_2 - t_1)$$

$$Q_b = \text{rm}$$

$$Q = cm(t_2 - t_1) + rm$$

Topish kerak: Q—?

Yechish: $Q = 4200\ \frac{J}{kg \cdot {}^0C} \cdot 0{,}001\ kg \cdot (100^0C - 0^0C) +$

$+ 2{,}3 \cdot \frac{10^6 \text{J}}{\text{kg}} \cdot 0{,}001\text{kg} = 420\text{J} + 2300\text{J} = 2720\text{J}$

Javob: $Q = 2720\text{J}$

Mustaqil bajarish uchun topshiriqlar

1. Mavzuni o‘qish va savollarga javob yozish.(119-bet)

2. Temperaturasi **100ºC** , massasi **2,5 kg** bo‘lgan suv bug‘i kondensatsiyalanganda qancha issiqlik miqdori ajraladi?